Perform the TG-DTA analysis of titanyl sulfate and citric acid and explain the result of analysis

TG-DTA Analysis of the Titanyl Sulfate and Citric Acid System

Abstract

This paper presents a comprehensive thermal analysis of the solution combustion synthesis system comprising titanyl sulfate ($TiOSO_4$) as the titanium precursor and citric acid ($C_6H_8O_7$) as the fuel. Thermogravimetric (TG) and differential thermal analysis (DTA) reveal a characteristic multi-stage decomposition pathway. The analysis identifies three primary thermal events: dehydration of the precursor and adsorbed water (below 200°C), decomposition of the titanyl-citrate complex and combustion (200–550°C), and crystallization of anatase TiO_2 accompanied by removal of residual carbonaceous species (above 550°C). The combustion reaction is highly exothermic and self-sustaining, enabling the formation of phase-pure anatase TiO_2 with crystallite sizes of approximately 14.8 nm under optimized conditions.

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1. Introduction

Thermogravimetric analysis (TGA) and differential thermal analysis (DTA) are indispensable techniques for understanding the thermal behavior of precursor systems in materials synthesis. In the solution combustion synthesis of titanium dioxide using titanyl sulfate ($TiOSO_4$) and citric acid, thermal analysis provides critical insights into the dehydration, decomposition, and phase formation stages that govern the final product characteristics. Understanding these thermal events is essential for optimizing synthesis parameters such as fuel-to-oxidizer ratio, ignition temperature, and calcination conditions to achieve the desired phase composition, crystallinity, and particle morphology .

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2. Materials and System Description

2.1 Titanyl Sulfate ($TiOSO_4$)

Titanyl sulfate, also known as titanium(IV) oxysulfate (CAS Number: 13825-74-6), is a white to yellowish, hygroscopic solid that is soluble in water. The anhydrous form has a molecular formula of $TiOSO_4$ with a molecular weight of 159.93 g/mol, while the hydrated form exists as $TiOSO_4 \cdot xH_2O$. The compound is a key intermediate in the sulfate process for producing titanium dioxide pigments and serves as a valuable precursor for synthesizing advanced titanium-based materials .

Under heating, titanium(IV) oxysulfate undergoes thermal decomposition, emitting toxic fumes of sulfur oxides (SOx). This decomposition process can be utilized to form TiO_2 particles .

2.2 Citric Acid ($C_6H_8O_7$)

Citric acid is a tricarboxylic acid that functions as an effective fuel in solution combustion synthesis. It serves two primary roles:

1. **Fuel-oxidizer redox reaction**: Citric acid acts as the reducing agent, providing electrons to reduce the titanyl species

2. **Metal complexation**: Citric acid acts as a tridentate chelating agent, forming stable titanyl-citrate complexes through coordination of its carboxylate and hydroxyl groups with Ti^{4+} ions

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3. TG-DTA Results and Analysis

3.1 Thermal Decomposition of Pure Citric Acid

The thermal degradation of citric acid provides a baseline for understanding its behavior in the $TiOSO_4$-citric acid system. DTA analysis of pure citric acid reveals three characteristic endothermic peaks at approximately 170°C, 185°C, and 210°C. These peaks correspond to:

- **170°C**: Onset of dehydration and partial decomposition of the hydrocarbon chain
- **185°C**: Further dehydration and initiation of decomposition
- **210°C**: Release of CO_2 and complete breakdown of the citrate structure

Under oxidizing conditions, the thermal degradation of citric acid occurs in two main stages. The first stage, extending until approximately 233°C, is associated with an 81% mass loss and corresponds to the release of H_2O and CO_2. The second stage, continuing until approximately 501°C, accounts for the remaining 19% mass loss. At the final decomposition temperature, the mass loss reaches 100%, indicating complete combustion of the organic acid .

The evolution of gaseous products during citric acid decomposition is characterized by the release of H_2O (indicated by m/z 17 and 18 signals) and CO_2 (indicated by m/z 12 and 44 signals). The first DTA endothermic peak with a minimum at 162°C is ascribed to dehydration and partial decomposition, while the endothermic peak at 213°C results from further dehydration and decomposition of the hydrocarbon chain with CO_2 release .

3.2 Thermal Behavior of Titanyl Sulfate

The thermal decomposition of titanium oxysulfate ($TiOSO_4 \cdot xH_2O$) is a multi-step process involving dehydration, formation of an anhydrous intermediate, decomposition of the sulfate group, and crystallization of titanium dioxide .

Table 1: Thermal Decomposition Characteristics of Titanyl Sulfate

Compound	Thermal Event	Temperature Range	Product(s)

| $TiOSO_4 \cdot H_2O$ | Loss of crystal water | ~310°C | Anhydrous $TiOSO_4$ |

| $TiOSO_4 \cdot H_2O$ | Liberation of SO_3 (decomposition) | ~485°C | Two-step decomposition observed in TG/DTG/DTA |

| $TiOSO_4$ (anhydrous) | Reversible phase transition | ~330°C | High-temperature phase |

| $TiOSO_4$ (anhydrous) | Decomposition to TiO_2 | ~525°C | Anatase TiO_2 |

The decomposition process typically begins with the loss of water molecules at temperatures below 300°C, followed by the formation of an amorphous intermediate phase. Subsequent heating leads to the evolution of sulfur oxides and the crystallization of TiO_2 in the anatase phase. This anatase phase can then transform into the more stable rutile phase at higher temperatures .

A study on the macromolecular (chitosan)·$(TiOSO_4)$n precursor using TGA/DTA revealed an exothermic peak around 260°C, confirming the decomposition of carbon chains to form graphite. Another exothermic peak around 560°C corresponds to the loss of additional carbonaceous residues. The formation of TiO_2 anatase was confirmed at approximately 400°C by Raman spectroscopy .

3.3 TG-DTA of the $TiOSO_4$-Citric Acid System

When titanyl sulfate and citric acid are combined in the solution combustion synthesis system, the thermal analysis reveals a characteristic sequence of events that governs the formation of TiO_2 nanoparticles.

The TGA curve of the precursor system typically shows multiple weight-loss stages corresponding to:

Stage 1: Dehydration (Below 200°C)

An endothermic peak is observed at approximately 100–150°C, corresponding to the evaporation of physically adsorbed water and water of crystallization from both the titanyl sulfate and citric acid. This stage accounts for approximately 10% weight loss .

Stage 2: Decomposition of Citrate Complex (200–500°C)

As the temperature increases, the titanyl-citrate complex undergoes thermal decomposition. This stage is characterized by:

- An exothermic peak around 260°C, corresponding to the decomposition of carbon chains
- Evolution of CO_2, H_2O, and SO_2 gases
- Onset of the combustion reaction

The decomposition of the citrate complex proceeds in a manner similar to pure citric acid, but the presence of titanyl species modifies the thermal behavior. Studies on combustion synthesis of aluminum titanate using citric acid show an exothermic peak at approximately 297°C corresponding to the decomposition of the metal complex, pyrolysis of organic groups, and oxide phase formation, resulting in a weight loss of approximately 75% .

Stage 3: Combustion and Phase Formation (500–550°C)

When the precursor mixture reaches the ignition temperature (optimally 550°C), the exothermic redox reaction between the titanyl species and citrate fuel initiates. This combustion is characterized by:

- Rapid evolution of large volumes of gases (CO_2, H_2O, SO_2)
- A highly exothermic and self-sustaining reaction
- Instantaneous rise in temperature
- Sharp and rapid weight change observed in the TG plot

The heat generated by the combustion reaction is sufficient to drive the crystallization of TiO_2 in the anatase phase. XRD analysis confirms that the combustion product at 550°C consists of phase-pure anatase TiO_2 with a crystallite size of approximately 14.8 nm .

Stage 4: Removal of Residual Carbon (Above 550°C)

An exothermic peak around 560°C corresponds to the loss of additional carbonaceous residues . This stage ensures the purity of the final TiO_2 product by eliminating unburnt carbon species that may have formed during the combustion.

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4. Interpretation of Thermal Events

4.1 Dehydration and Complex Formation

The initial heating of the precursor solution leads to the formation of a titanyl-citrate complex through coordination of the citrate carboxylate groups with Ti^{4+} ions. This complexation is essential for maintaining molecular-level homogeneity and preventing premature hydrolysis of titanium species. The dehydration stage removes water, concentrating the complex and preparing it for thermal decomposition .

4.2 Combustion Mechanism

The combustion reaction between titanyl sulfate and citric acid is driven by the highly exothermic oxidation of the citrate fuel. The redox reaction can be represented as:

$TiOSO_4 + C_6H_8O_7 + O_2 \rightarrow TiO_2$ (anatase) $+ CO_2 + H_2O + SO_2$

The actual reaction pathway involves complex multi-step processes, including:

- Oxidation of carboxylate groups

- Evolution of reducing gases (CO, H_2, hydrocarbons)

- Reduction of titanyl species

- Crystallization of TiO_2

The presence of HNO_3 in some synthesis routes plays a key role as an oxidizer, with the oxidation of the citrate precursor by HNO_3 accompanied by the evolution of CO_2, NO_2, and water vapor. This gas evolution helps produce a fine-grained structure .

4.3 Phase Formation and Crystallization

The high temperature and rapid heating rate during combustion promote the nucleation and growth of anatase TiO_2 crystallites. The large volume of gases evolved during combustion:

- Prevents particle agglomeration

- Limits crystallite growth to the nanoscale (14.8 nm)

- Creates a porous, high-surface-area product

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5. Influence of Synthesis Parameters on Thermal Behavior

5.1 Effect of Fuel-to-Oxidizer Ratio

The molar ratio of citric acid to titanyl sulfate significantly affects the combustion behavior and product properties. At the optimal Ti-to-citrate ratio of 9:15 (approximately 0.6:1 molar ratio), the redox stoichiometry is balanced, ensuring complete combustion and minimal residual carbon. The combustion temperature and exothermicity are maximized at this stoichiometric ratio .

5.2 Effect of pH

The pH of the precursor solution influences the speciation of titanyl-citrate complexes and the stability of the solution. Optimal conditions require a pH greater than 5 to ensure complete dissolution and complexation of titanium species. At lower pH values, the hydrolysis of titanium ions leads to the precipitation of titanium hydroxide, compromising the homogeneity of the precursor and altering the thermal behavior .

5.3 Effect of Ignition Temperature

The ignition temperature is a critical parameter that determines the initiation and propagation of the combustion front. An ignition temperature of 550°C has been identified as optimal for the titanyl sulfate-citric acid system . The thermal decomposition of pure titanyl sulfate shows that decomposition to TiO_2 occurs at approximately 525°C , which aligns well with the observed ignition temperature.

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6. Summary and Conclusions

The TG-DTA analysis of the titanyl sulfate and citric acid system reveals a well-defined thermal pathway that governs the formation of nanocrystalline TiO_2:

Table 2: Summary of TG-DTA Events in the $TiOSO_4$-Citric Acid System

Stage	Temperature	Thermal Event	DTA Signal	Weight Loss	Product
1	< 200°C	Dehydration of adsorbed water and crystal water	Endothermic (100–150°C)	~10%	Dehydrated precursor

| 2 | 200–260°C | Decomposition of citrate complex and carbon chain breakdown | Exothermic (~260°C) | ~75% | Amorphous Ti-O-C species |

| 3 | 260–550°C | Combustion reaction; SO_3 liberation | Exothermic (combustion) | ~15% | Anatase TiO_2 (14.8 nm) |

| 4 | > 550°C | Removal of residual carbonaceous species | Exothermic (~560°C) | Minimal | Purified anatase TiO_2 |

The key findings from the thermal analysis are:

1. **Dual role of citric acid**: Citric acid functions both as a fuel for the combustion reaction and as a complexing agent that stabilizes titanyl species, ensuring molecular-level homogeneity.

2. **Self-sustaining combustion**: The exothermic decomposition of the titanyl-citrate complex generates sufficient heat to propagate the combustion reaction without external energy input after ignition.

3. **Low-temperature crystallization**: The combustion heat enables crystallization of anatase TiO_2 at temperatures as low as 400–550°C, significantly lower than conventional synthesis routes.

4. **Nanosized product**: The rapid heating and gas evolution during combustion limit crystallite growth, yielding nanoparticles with a crystallite size of approximately 14.8 nm .

5. **Phase control**: The thermal conditions during combustion favor the formation of anatase TiO_2. Higher temperatures or extended calcination would be required to induce transformation to the rutile phase.

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